

DSA 8420 Spring 2025

Project 2

Due April 15, 2025

In this project, we solve an assignment problem using the Python package *scipy.optimize*. *Scipy.optimize* provides functions for minimizing (or maximizing) objective functions, possibly subject to constraints. It includes solvers for nonlinear problems, linear programming, constrained and nonlinear least-squares, root finding, and curve fitting. In this project, we use two functions in the package to solve the assignment problem in two ways — as a general linear program and as an assignment problem.

Installation. You need to install two packages (Numpy and Scipy) in our Python environment. In your terminal or Anaconda prompt,

- Activate our environment:
`conda activate opt.in.python`
- Install Numpy:
`conda install numpy`
- Install Scipy:
`conda install scipy`

Task 1. (35 points) Let $C = [c_{ij}]$ be the cost matrix of an assignment problem. The problem can be modeled as a linear program:

$$\begin{aligned} \min \quad & \sum_{i=1}^n \sum_{j=1}^n c_{ij} x_{ij} \\ \text{s.t.} \quad & \sum_{i=1}^n x_{ij} = 1 && \text{for } j = 1, \dots, n \\ & \sum_{j=1}^n x_{ij} = 1 && \text{for } i = 1, \dots, n \\ & x_{ij} \geq 0 && \text{for } i, j = 1, \dots, n \end{aligned}$$

Define `A_eq` and `b_eq` in the first part of the Jupyter Notebook *assignment.ipynb*. Your code should work for a general n . Use the following documents and/or other online resources for help.

- `scipy.optimize.linprog`:
<https://docs.scipy.org/doc/scipy/reference/generated/scipy.optimize.linprog.html>
- Numpy basics:
https://numpy.org/doc/stable/user/absolute_beginners.html

Task 2. (15 points) Instead of solving the assignment problem as a linear program, the function `scipy.optimize.linear_sum_assignment()` solves the assignment problem directly. Read the following document and fill in the cell in the second part of the Jupyter Notebook *assignment.ipynb*. Your code should solve the problem with the cost matrix `cost` and record the elapsed time.

- https://docs.scipy.org/doc/scipy/reference/generated/scipy.optimize.linear_sum_assignment.html

If coded correctly, the minimum costs returned by the two methods should be identical (up to some tiny numerical error).

Task 3. (10 points) Run the file with different number of jobs n . (You can click on Restart & Run All under the Kernel tab.) Record the elapsed time for both methods. Fill in the following table.

n	10	20	50	100	200
LP time (sec)					
assignment time (sec)					

Which method is faster?

Submission. Your submission should include two parts.

- Part I: Fill in the blank cells in ‘`assignment.ipynb`’ and submit the file **to Canvas**.
- Part II: Put down your answer to Task 3 (and any other comments) in a pdf file and submit it **to Gradescope**.